

YUWEI ZHUO

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EDUCATION

University of Michigan, Michigan, The United States

Expected May 2027

Bachelor of Engineering in Industrial & Operations Engineering (IOE)

GPA: 3.97

Coursework: Optimization, Data Analysis, Business Analytics, Markov Processes, Statistical Modeling, Python Programming University of Michigan Solar Car Team *Operations Team Member*

- Coordinated operational workflows involving route planning, procurement, logistics, and transportation for a competition-grade solar vehicle
- Supported resource allocation initiatives contributing to partnerships totaling over \$2.4M

Shanghai Jiao Tong University (SJTU), Shanghai, China

Expected July 2027

Bachelor of Engineering in Electrical & Computer Engineering (ECE)

Coursework: Data Structures & Algorithms, C/C++ Programming, Deep Learning, Signals & Systems, Digital Logic Design

WORK EXPERIENCE

Sensetime Group Inc., *AI Product Manager Intern*

May. 2026 - Aug. 2026

- Developed a quality management AI data analysis platform for Shanghai Electric Group Company, improving their efficiency ...
- Designed a ...

SJTU GAIR Lab, *Research Assistant*

Nov. 2024 - Jan. 2025

- Developed and iteratively optimized prompting frameworks for an autonomous problem-solving agent focused on Chinese Gaokao-style multi-subject reasoning tasks, improving overall task accuracy by nearly 50%
- Constructed benchmark datasets from publicly available Gaokao problems and designed subject-specific evaluation workflows to systematically assess agent performance across diverse reasoning tasks
- Designed adaptive prompting strategies for different question types and conducted iterative error analysis to identify reasoning bottlenecks and improve response quality
- Built an automated evaluation pipeline using a secondary grading agent with standardized scoring rubrics, enabling scalable and reproducible assessment of generated solutions

RESEARCH EXPERIENCE

Kidney Transplant Queuing System Design, *Research Assistant*

Jan. 2026 - Present

- Conducted literature reviews and evidence synthesis on kidney donor viability, transplant allocation systems, and metabolic risk factors affecting donor kidney utilization
- Analyzed historical kidney transplant datasets to investigate relationships between donor profiles, transplant outcomes, and organ discard patterns under varying clinical conditions
- Evaluated the impact of obesity, prediabetes, and diabetes-related factors on donor kidney quality and transplant feasibility to support healthcare system decision-making under uncertainty
- Prepared analytical summary reports integrating clinical studies, registry-based findings, and operational considerations for transplant system modeling research

Electric Market Trading System Design, *Researcher*

March. 2026 – Present

- Developed analytical frameworks for demand response and electricity market decision-making using price elasticity estimation and counterfactual pricing methodologies
- Built data-driven forecasting pipelines integrating load, renewable generation, and market pricing signals to analyze industrial user behavior under dynamic market conditions
- Explored optimization strategies for energy trading and demand-side management under uncertainty through scenario-based analysis and predictive modeling techniques

SKILLS

- Programming & Analytics: Python, SQL, C++, Java, C
- Data Analysis & Modeling: pandas, statistical analysis, simulation, numerical optimization
- Research & Visualization: $\text{\LaTeX}$, matplotlib, Microsoft Excel, Power BI, PowerPoint